

DATA CLEANING

Learning Material for Statistical Officer

1. Introduction to Data Cleaning

Data cleaning is the process of identifying, correcting, removing, or replacing incorrect, incomplete, duplicate, and inconsistent data in a dataset. Data cleaning is an important step in data analysis because the quality of input data directly affects the accuracy of analysis and results.

A dataset may contain:

- Missing values
- Duplicate records
- Incorrect values
- Spelling mistakes
- Inconsistent formats
- Outliers

2. Why is Data Cleaning Important?

Poor-quality data can lead to incorrect conclusions. Data cleaning improves data quality, reduces errors, improves accuracy, removes unnecessary duplicate information, makes data easier to analyze, and supports better decision-making.

For statistical officers, clean data is especially important because statistical reports and decisions depend on the quality of collected data.

3. Common Data Quality Problems

A dataset can contain several types of problems.

3.1 Missing Values

A missing value occurs when information is not available for a particular record.

Employee	Age	Department
Ravi	25	Statistics
Kumar		Statistics
Anil	28	IT

Common approaches for handling missing values include removing the record, replacing the value with the mean, median, or mode. The appropriate method depends on the type and importance of the data.

4. Duplicate Records

A duplicate record is a repeated entry in a dataset. Duplicate records should be identified and removed when they represent the same entity. Removing duplicates prevents information from being counted

more than once.

ID	Name	Department
101	Ravi	Statistics
102	Kumar	IT
101	Ravi	Statistics

5. Incorrect Data

Incorrect data contains values that do not represent the actual information. For example, an age of 250 may be an incorrect value. A spelling such as 'Malee' may also indicate an error. Incorrect values should be checked and corrected using reliable source information whenever possible.

6. Inconsistent Data Formats

Data may represent the same information using different formats. For example:

Karnataka
karnataka
KARNATAKA

These values can be standardized as 'Karnataka'. Standardization makes searching, grouping, and analysis easier.

7. Handling Outliers

An outlier is a value that is significantly different from other values in a dataset.

45, 48, 47, 50, 46, 49, 250

Here, 250 may be an outlier if these values represent ages. Outliers should not always be removed automatically. They should first be investigated to determine whether they are data entry errors, measurement errors, or genuine unusual observations.

8. Data Standardization

Data standardization is the process of converting data into a consistent format.

State names before standardization:

KA
Karnataka
KARNATAKA

A consistent representation can be:

Karnataka
Karnataka
Karnataka

Date values can also be standardized. For example, 12/01/2026, 2026-01-12, and 12-Jan-2026 may be converted to a selected standard such as 2026-01-12.

9. Basic Data Cleaning Process

A basic data cleaning process can be performed using the following steps:

Raw Dataset → Identify Data Problems → Handle Missing Values → Remove Duplicates
→ Correct Incorrect Values → Standardize Data → Check Outliers → Validate
Cleaned Data → Clean Dataset

10. Data Validation

Data validation is the process of checking whether cleaned data satisfies required rules and conditions.

- Age should be within a reasonable range.
- Required fields should not be empty.
- State names should follow a standard format.
- IDs should be unique.
- Numerical values should contain valid numbers.

11. Example

ID	Name	State	Age
101	Ravi	Karnataka	25
102	Kumar	karnataka	
101	Ravi	Karnataka	25
103	Anil	KARNATAKA	250

Problems in this dataset include a duplicate record for Ravi, a missing age for Kumar, inconsistent state names, and a possible incorrect age for Anil.

After cleaning, the dataset could be represented consistently. The replacement values for missing or incorrect data must be determined using an appropriate method or reliable source in a real application.

12. Key Points

- **Missing values** → Information is unavailable.
- **Duplicate records** → Same information appears more than once.
- **Incorrect data** → Values contain errors.
- **Inconsistent data** → Same information uses different formats.
- **Outliers** → Values that are unusually different.
- **Standardization** → Converting data into a consistent format.
- **Validation** → Checking whether cleaned data meets required rules.

Conclusion

Data cleaning is an essential part of the data analysis process. It improves the quality, consistency, and reliability of data. Properly cleaned data helps statistical officers perform accurate analysis and produce reliable statistical information.